Lecture 2 (Review of Linear Algebra and LP Result)

January 22, 2026

We will continue our review of linear algebra concepts, then we will prove a big result for LPs.

Index Notation

When working with linear algebra proofs, it is easier to use index notation. Consider the matrix,

$$A = \begin{pmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nm} \end{pmatrix}$$

If we want to refer the to the entries of A, we often use the notation $A_{ij} = a_{ij}$. If a capital letter is a matrix, then an indexed capital letter is an entry of A. The book uses similar notation. Instead the book uses the notation $[A]_{ij}$. This is to make it clear that $[A]_{ij}$ is an element, rather than another matrix. Generally, I prefer the first notation, but I may use both. In many fields people will use one or the other depending on context. For vectors, we can do something similar:

$$\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$$

yields $\vec{a}_i = [\vec{a}]_i = a_i$. Often we will be lazy and drop the arrow above the vector when indexing, especially if it is clear by context. The final shorthand we will often use is summation over repeated indices. This is to avoid writing many summations.

$$\sum_i x_i A_{ij} = x_i A_{ij}$$

This shorthand is called **Einstein Summation Notation**. Let's give a few examples of this notation in practice.

Ex 1)

Consider the matrix multiplication

$$C = AB$$

If we were to explicitly write the matrix multiplication, this would involve a mess of column and row vectors. Instead, we can denote this as

$$C_{ij} = A_{ik} B_{kj}$$

This product tells us how to find each entry of C. (Remember here we are dropping the summation, since we are summing over all k). For proofs involving repeated multiplication, we can often use this to save a lot of writing. Consider the following product:

$$A = BCDEFG$$

If we wished to write the product below explicitly with summations, we would have the long expression below:

$$A_{ij} = \sum_o \sum_n \sum_m \sum_l \sum_k B_{ik} C_{kl} D_{lm} E_{mn} F_{no} G_{oj}$$

But, we can save writing and write this instead:

$$A_{ij} = B_{ik} C_{kl} D_{lm} E_{mn} F_{no} G_{oj}$$

When writing proofs, this will save a lot of writing over many steps.

Ex 2)

Consider the transpose operation. If matrix

$$A = \begin{pmatrix} a_{11} & \dots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nm} \end{pmatrix}$$

then

$$A^T = \begin{pmatrix} a_{11} & \dots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \dots & a_{mn} \end{pmatrix}$$

The idea here is that we swap the rows and columns of the matrix. This shows up in our index notation by defining:

$$A^T_{ij} = A_{ji}$$

Notice the key idea of swapping rows and columns is defined formally without the need to write out the full matrix.

Some Proofs With Index Notation

Let's do a couple of proofs using the index notation for practice. I encourage you to try these proofs on your own before checking how I wrote them. There may be multiple ways of proving these results, and the practice is always good.

Ex 3)

Show

$$(A+B)^T = A^T + B^T$$

Proof

$$(A+B)^T_{ij} = (A+B)_{ji} = A_{ji} + B_{ji} = A^T_{ij} + B^T_{ij}$$

Since each the statement above holds for all i, j then it is true for the matrices, thus

$$(A+B)^T = A^T + B^T$$

Ex 4)

Recall the **trace** is the sum

$$\mathrm{tr}\,(A) = \sum_i A_{ii}$$

(I am writing this explicitly to avoid any ambiguity). Show that

$$\mathrm{tr}\,(AB) = \mathrm{tr}\,(BA)$$

Proof

$$\mathrm{tr}\,(AB) = \sum_i [AB]_{ii}$$

and using index notation (remember the repeated index means to sum over the repeated index!), we see

$$\sum_i [AB]_{ii} = \sum_i A_{ik}B_{ki} = \sum_i B_{ki}A_{ik} = \sum_k B_{ki}A_{ik} = \sum_k [BA]_{kk} = \mathrm{tr}(BA)$$

Here, I deliberately used the square bracket notation and the summations mixed without the summations. You can use whichever notation as long as it is clear. Normally, we pick one and stick to it. On occasion a sum is added for clarity. Also note that this result doesn't imply that AB has the same dimensions of BA. By dropping the sums, we can see which quantities (like trace) are preserved without having to keep the dimensions straight.

Ex 5)

Show

$$\mathrm{tr}\left(A^T\right) = \mathrm{tr}\left(A\right)$$

Proof

$$A^T_{ij} = A_{ji}$$

Thus

$$\mathrm{tr}\left(A^T\right) = \sum_j A_{jj} = \sum_i A_{ii} = \mathrm{tr}\left(A\right)$$

Dot Products

One important operation is the **dot product**. This is defined as

$$\vec{a} \cdot \vec{b} = a_i b_i = \sum_i a_i b_i$$

(I wrote the summation explicitly and dropped it to emphasize the repeated summation rule). The dot product is a special case of a more general object called an inner product (which is written as $\left\langle \vec{a}, \vec{b} \right\rangle$ and is subject to a few axioms). As such, you will occasionally see proofs involving $\left\langle \vec{a}, \vec{b} \right\rangle$. These are certainly valid for dot products, and if the inner product isn't specified it is almost always going to be the dot product. Let's show some properties of the dot product.

Properties

Let $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and $c \in \mathbb{R}$

The dot product obeys the following properties:

1) Positive Definite

$$\vec{a} \cdot \vec{a} \geq 0$$

and

$$\vec{a} \cdot \vec{a} = 0$$

if and only if $\vec{a} = 0$.

2) (Conjugate) Symmetry

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

3) Linearity

$$\left(\vec{a} + \vec{b}\right) \cdot \vec{c} = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$$

and

$$(c\vec{a}) \cdot \vec{b} = c\left(\vec{a} \cdot \vec{b}\right) + \vec{a} \cdot \left(c\vec{b}\right)$$

Proof

Before reading through the proof, I encourage you to try proving this yourself from the index notation.

1)

$$\vec{a} \cdot \vec{a} = a_i a_i = \sum_i a_i^2 \geq 0$$

as each a_i is real. Further, $\sum_i a_i^2 = 0$ implies each $a_i = 0$. If each $a_i = 0$, then $\vec{a} = 0$.

2)

$$\vec{a} \cdot \vec{b} = a_i b_i = b_i a_i = \vec{b} \cdot \vec{a}$$

3)

$$\left(\vec{a} + \vec{b}\right) \cdot \vec{c} = (a_i + b_i)\, c_i = a_i c_i + b_i c_i = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$$

and

$$(ca_i)\, b_i = c\,(a_i b_i) = a_i\,(cb_i)$$

as each c, a_i, b_i is a real number, thus

$$(c\vec{a}) \cdot \vec{b} = c\left(\vec{a} \cdot \vec{b}\right) + \vec{a} \cdot \left(c\vec{b}\right)$$

In general, an inner product any binary operation that satisfies properties **1)**, **2)**, and **3)**. Also note that these properties also imply

$$\vec{a} \cdot \left(\vec{b} + \vec{c}\right) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

Combining this with linearity is called **bilinearity**.

Length of a Vector

We can find the **length** of a vector using Pythagorean theorem. This is usually denoted as

$$\|\vec{a}\| = \sqrt{a_i a_i} = \sqrt{\sum_i a_i^2}$$

Notice that this is equivalent to saying

$$\|\vec{a}\| = \sqrt{\vec{a} \cdot \vec{a}}$$

This operation is a special case of an object called a **norm**. We will not worry too much about the general definition. For our purposes, we will just use length. Now, let's show the key result:

Cosine Similarity

Let $\vec{a}, \vec{b} \in \mathbb{R}$. Then if θ is the angle between the vectors, the following holds:

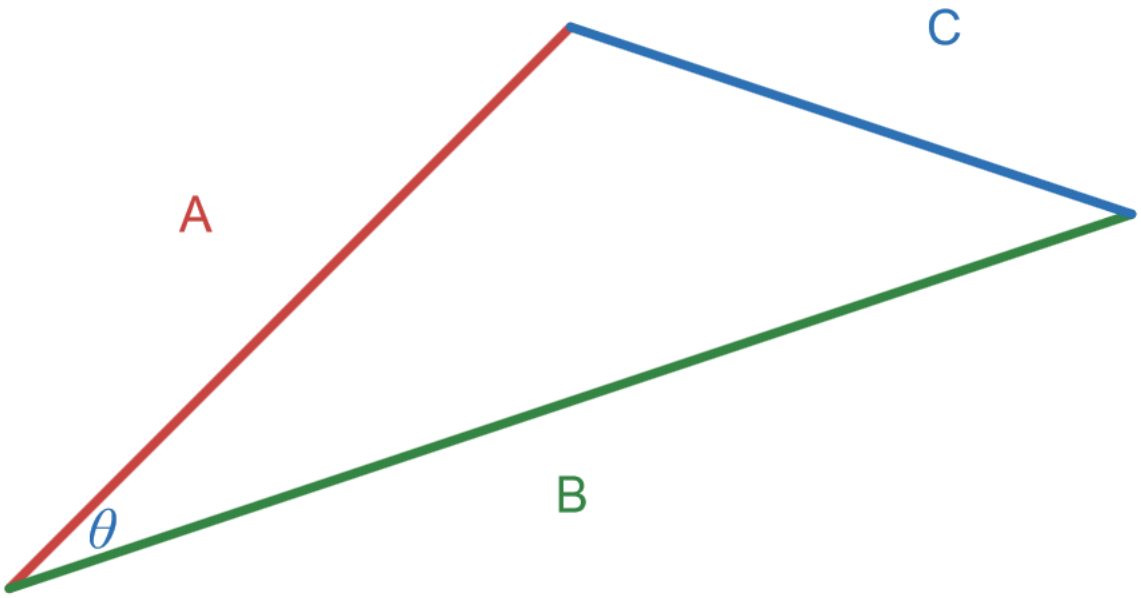
$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \left\|\vec{b}\right\| \cos(\theta)$$

Proof

Recall the Law of Cosines:

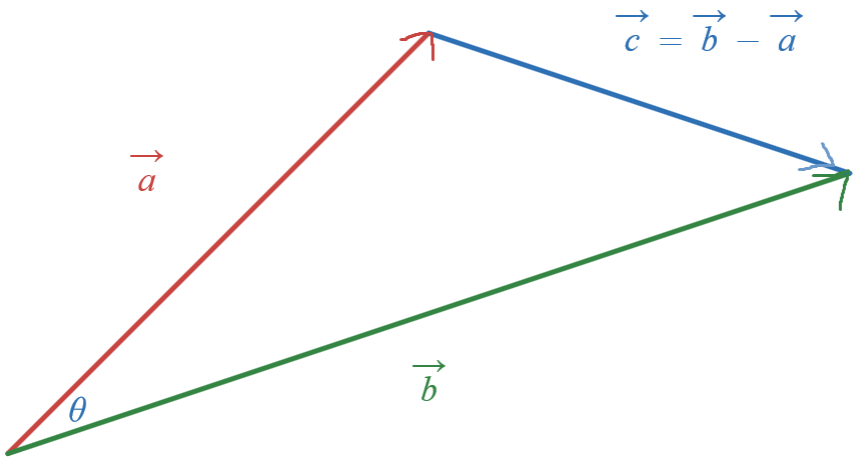
$$C^2 = A^2 + B^2 - 2AB\cos(\theta)$$

where θ is the angle opposite of C.



We can rewrite this in terms of vectors. If we let A, B be replaced with $\vec{a}, \vec{b}$ respectively (where $\vec{a}, \vec{b}$ start at the origin), and we replace C with $\vec{c} = \vec{b} - \vec{a}$), then we see the equivalent statement of the law of cosines:

$$\|\vec{c}\|^2 = \|\vec{a}\|^2 + \left\|\vec{b}\right\|^2 - 2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$



Now, we can prove the desired result using the properties of the dot product:

$$\|\vec{c}\|^2 = \|\vec{a}\|^2 + \left\|\vec{b}\right\|^2 - 2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

$$\left\|\vec{b} - \vec{a}\right\|^2 = \|\vec{a}\|^2 + \left\|\vec{b}\right\|^2 - 2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

$$\left(\vec{b} - \vec{a}\right) \cdot \left(\vec{b} - \vec{a}\right) = \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} - 2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

By linearity

$$\vec{b} \cdot \vec{b} - \vec{b} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{a} = \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} - 2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

By symmetry

$$\vec{b} \cdot \vec{b} - \vec{a} \cdot b - \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{a} = \vec{a} \cdot \vec{a} + \vec{b} \cdot \vec{b} - 2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

Combine like terms and simplify

$$-2\vec{a} \cdot \vec{b} = -2\|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\left\|\vec{b}\right\|\cos(\theta)$$

A Result About LPs

Suppose an optimization problem is a linear program. The **feasible region** of a linear program is the region containing all points that satisfy all constraints. If the collection of inequalities yields a contradictory result, the feasible region is said to be **empty** and the linear program is said to be **infeasible**. This LP will have no solution If the feasible region is **unbounded**, the LP is said to be **unbounded**. Some unbounded LPs have a solution. We will elaborate more on these two cases next class. Now, we can state our first LP theorem:

Theorem

If an LP has a bounded, non-empty feasible region, then an optimal solution will exist on a vertex of the feasible region.

Proof

In an LP, the objective function is of the form:

$$f(\vec{x}) = c_0 + \sum_i c_i x_i = c_0 + c_i x_i$$

Recall from calculus 3, that an optimal solution must have a gradient of 0 or lie on the boundary of the feasible region (assuming the gradient exists everywhere in the feasible region). We can take the gradient of f, to see

$$\nabla f = (c_1, c_2, \ldots, c_n) = c_i$$

(The c_i is in index notation, the first vector is the standard calculus notation). Now, we consider two cases:
Case 1)

$$\nabla f = \vec{0}$$

This implies

$$c_i = 0$$

so

$$f(\vec{x}) = c_0$$

A constant function is optimal everywhere, so an optimal solution will exist on the vertex (in fact, every point in the feasible region will be optimal). For the remainder of the proof, we will consider the more interesting case.
Case 2)

$$\nabla f \neq \vec{0}$$

In this case, the gradient will never be $\vec{0}$ at any points of the feasible region. This is because the gradient is constant for a linear function. This implies that the optimal solution must lie on the boundary of the feasible region. If the feasible region is a k dimensional polytope, then this implies that the solution will lie in the $k-1$ dimensional boundary. This boundary will consist of $k-1$ dimensional shapes joined together. If we restrict the objective function to each of these shapes, we can use the same argument to show that the an optimal solution must be on one of the $k-2$ dimensional boundaries of these shapes (after all, the gradient is still non-zero on each of the $k-1$ dimensional shapes). We repeat this argument until we are left with the vertices of the polytope. Therefore, an optimal solution must lie on a vertex of the polytope.

Final Thoughts

Case 2) does not imply that all optimal solutions lie on the vertices of the feasible region. Just that an optimal solution will. Consider the following example:

$$\text{maximize } f(x, y) = y$$

Subject to

$$0 \leq x \leq 3$$

$$0 \leq y \leq 2$$

Notice that the feasible region is a rectangle with vertices

$$(0,0), (0,2), (3,0), (3,2)$$

and that

$$f(0,0) = f(3,0) = 0$$

$$f(0,2) = f(3,2) = 2$$

But we also have

$$f(1,2) = 2$$

and $(1,2)$ is not a vertex of this rectangle. All we know is that an optimal solution will lie on a vertex of the feasible region. Not that all optimal solutions do.

